Equilibrium Existence, Uniqueness and Stability

230333 Microeconomics 3 (CentER) – Part II
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Introduction

In this section we will discuss three topics:

- **Existence:** Under which assumptions we can be guaranteed that an equilibrium will exist?
  - We will do two proofs, where one deals with the complication of strongly monotonic preferences with zero prices.
- **Uniqueness:** Under which conditions can we be guaranteed that an equilibrium will be unique?
- **Stability:** Under which conditions is an equilibrium stable?
  - If the economy is pushed away from equilibrium (e.g. from a shock), will it adjust back?

Equilibrium in Pure Exchange Economies

- A pure exchange economy is a special case of the general case with $J = 1$ and $Y_1 = -\mathbb{R}^L_+$ (*free disposal*).
- If $\bar{\boldsymbol{\omega}} \gg \mathbf{0}$ and each consumer i has continuous, strictly convex and locally nonsatiated preferences, the equilibrium definition can be restated as:

Definition

$\left(\boldsymbol{x}^\star, \boldsymbol{y}_1^\star\right)$ and $\boldsymbol{p} \in \mathbb{R}^L$ constitute a Walrasian equilibrium in a pure exchange economy iff:

(i) $\boldsymbol{y}_1^\star \leq \mathbf{0}$, $\boldsymbol{p} \cdot \boldsymbol{y}_1^\star = 0$ and $\boldsymbol{p} \geq \mathbf{0}$ (profit maximization).

(ii) $\boldsymbol{x}_i^\star = \boldsymbol{x}_i\left(\boldsymbol{p}, \boldsymbol{p} \cdot \boldsymbol{\omega}_i\right)$ for all i (utility maximization).

(iii) $\sum_{i=1}^I \boldsymbol{x}_i^\star = \sum_{i=1}^I \boldsymbol{\omega}_i + \boldsymbol{y}_1^\star$ (market clearing).

Excess Demand

- The *excess demand function of consumer i* is:

$$\boldsymbol{z}_i(\boldsymbol{p}) = \boldsymbol{x}_i(\boldsymbol{p}, \boldsymbol{p} \cdot \boldsymbol{\omega}_i) - \boldsymbol{\omega}_i$$

- The *aggregate excess demand function* of the economy is:

$$\boldsymbol{z}(\boldsymbol{p}) = \sum_{i=1}^{I} \boldsymbol{z}_i(\boldsymbol{p})$$

- In a pure exchange economy in which preferences are continuous, strictly convex and locally nonsatiated, $\boldsymbol{p} \geq \boldsymbol{0}$ is a Walrasian equilibrium price vector iff $\boldsymbol{z}(\boldsymbol{p}) \leq \boldsymbol{0}$.
  - $\boldsymbol{y}_1^{\star} = \boldsymbol{z}(\boldsymbol{p})$ is profit-maximizing, because $\boldsymbol{p} \cdot \boldsymbol{z}(\boldsymbol{p}) = 0$.
    - $\boldsymbol{p} \cdot \boldsymbol{z}_i(\boldsymbol{p}) = 0 \ \forall i$ by Walras' law (LNS), so $\sum_{i=1}^{I} \boldsymbol{p} \cdot \boldsymbol{z}_i(\boldsymbol{p}) = 0$.

Proof of Existence

Proposition

Suppose that $\boldsymbol{z}(\boldsymbol{p})$ is a function defined for all nonzero, nonnegative price vectors $\boldsymbol{p} \in \mathbb{R}_+^L$ and satisfies continuity, homogeneity of degree zero and Walras' law. Then there is a price vector $\boldsymbol{p}^\star$ such that $\boldsymbol{z}\left(\boldsymbol{p}^\star\right) \leq \mathbf{0}$.

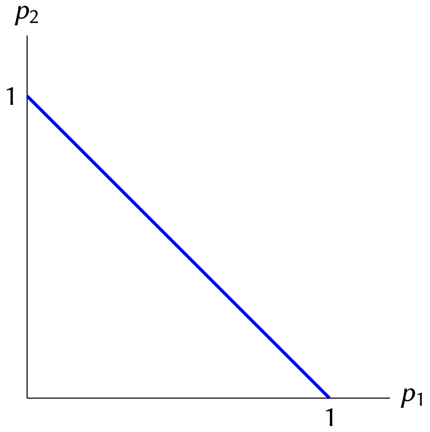
- Because of homogeneity of degree zero, we can normalize prices to the unit simplex:

$$\Delta = \left\{ \boldsymbol{p} \in \mathbb{R}_+^L : \sum_{\ell=1}^{L} p_\ell = 1 \right\}$$

- Δ is compact (closed and bounded) and convex.

Unit Simplex with $L = 2$

With $L = 2$, the unit simplex is given by the line $p_2 = 1 - p_1$, for $p_1 \in [0, 1]$.



Proof of Existence

- Define the function $f : \Delta \to \Delta$:

$$\{f_\ell(\boldsymbol{p})\}_{\ell=1}^{L} = \left\{ \frac{p_\ell + \max\{z_\ell(\boldsymbol{p}), 0\}}{1 + \sum_{k=1}^{L} \max\{z_k(\boldsymbol{p}), 0\}} \right\}_{\ell=1}^{L}$$

- Because $z_\ell(\boldsymbol{p})$ is continuous $\forall \ell$ and the denominator is bounded away from zero, f is continuous. See notes on Canvas for a formal ε-δ proof of continuity.
- f is a continuous function mapping a compact convex set to itself: Brouwer can be applied.
- By Brouwer's fixed-point theorem, $\exists \boldsymbol{p}^\star \in \Delta$ s.t. $\boldsymbol{p}^\star = f(\boldsymbol{p}^\star)$.

$$\underbrace{0 = \boldsymbol{p}^\star \cdot \boldsymbol{z}(\boldsymbol{p}^\star)}_{\text{Walras' law}} = f(\boldsymbol{p}^\star) \cdot \boldsymbol{z}(\boldsymbol{p}^\star) = \frac{\sum_{\ell=1}^{L} \left(p_\ell + \max\left\{z_\ell(\boldsymbol{p}^\star), 0\right\}\right) z_\ell(\boldsymbol{p}^\star)}{1 + \sum_{k=1}^{L} \max\{z_k(\boldsymbol{p}^\star), 0\}}$$

- Therefore $\sum_{\ell=1}^{L} \max\left\{z_\ell(\boldsymbol{p}^\star), 0\right\} z_\ell(\boldsymbol{p}^\star) = 0$, so $\boldsymbol{z}(\boldsymbol{p}^\star) \leq \boldsymbol{0}$.

Strongly Monotone Preferences

- The previous proof works when demand is continuous over all nonzero, nonnegative prices.
- However, if preferences are strongly monotone, demand is infinite at zero prices
  - This occurs at the boundary of the simplex.
- We will now adapt the proof to handle this case.

Properties of the Aggregate Excess Demand Function

Suppose that, for every consumer i, $X_i = \mathbb{R}_+^L$ and $\succeq_i$ is continuous, strictly convex, and strongly monotone. Suppose also that $\bar{\boldsymbol{\omega}} \gg \mathbf{0}$. Then the aggregate excess demand function, defined for all price vectors $\boldsymbol{p} \gg \mathbf{0}$ satisfies:

(i) $\boldsymbol{z}(\cdot)$ is continuous

(ii) $\boldsymbol{z}(\cdot)$ is homogenous of degree zero.

(iii) $\boldsymbol{p} \cdot \boldsymbol{z}(\boldsymbol{p}) = 0$ for all $\boldsymbol{p}$ (Walras' law)

(iv) There is an $s > 0$ such that $z_\ell(\boldsymbol{p}) > -s$ for every commodity ℓ and all $\boldsymbol{p}$.

(v) If $\boldsymbol{p}^n$ is a sequence of price vectors converging to $\boldsymbol{p} \neq \mathbf{0}$ and $p_\ell = 0$ for some ℓ, then $z_\ell(\boldsymbol{p}^n) \to \infty$.

- In the limit as the price of a good goes to zero, there is at least one consumer with positive wealth whose demand approaches infinity of the good.

Existence of Equilibria With Strongly Monotone Preferences

In a pure exchange economy in which consumer preferences are continuous, strictly convex, and strongly monotone, $\boldsymbol{p} \gg \mathbf{0}$ is a Walrasian equilibrium price vector if and only if:

$$\boldsymbol{z}(\boldsymbol{p}) = \mathbf{0}$$

Proposition

Suppose that $\boldsymbol{z}(\boldsymbol{p})$ is a function defined for all $\boldsymbol{p} \in \mathbb{R}^L_{++}$ satisfying conditions (i)-(v) on the previous slide. Then the system of equations $\boldsymbol{z}(\boldsymbol{p}) = \mathbf{0}$ has a solution. Hence, a Walrasian equilibrium exists in any pure exchange economy in which $\bar{\boldsymbol{\omega}} \gg \mathbf{0}$ and every consumer has continuous, strictly convex and strongly monotone preferences.

Unit Simplex

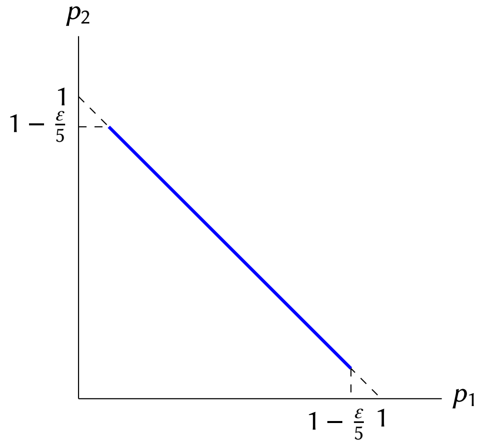
We define a variation on the unit simplex from the last proof.
For a fixed $\varepsilon \in (0, 1)$:

$$\Delta_\varepsilon = \left\{ \boldsymbol{p} : \sum_{\ell=1}^{L} p_\ell = 1 \text{ and } p_\ell \geq \frac{\varepsilon}{1 + 2L} \forall \ell \right\}$$

- Δ_ε is compact (closed and bounded).
- Δ_ε is convex.
- Δ_ε non-empty:
  - $p_\ell = \frac{1}{L}, \forall \ell$ is an element for any $\varepsilon \in (0, 1)$, because $\sum_{\ell=1}^{L} p_\ell = 1$ and $\frac{1+2L}{L} > \varepsilon$ for $\varepsilon \in (0, 1)$.
- Later we will let $\varepsilon \to 0$.

Δ_ε with $L = 2$

$$\Delta_\varepsilon = \left\{ \boldsymbol{p} : \sum_{\ell=1}^{L} p_\ell = 1 \text{ and } p_\ell \geq \frac{\varepsilon}{1+2L} \forall \ell \right\}$$



Fixed Point Function

Define for each $\boldsymbol{p} \in \Delta_\varepsilon$ a function $\boldsymbol{f}(\boldsymbol{p}) = \{f_\ell(\boldsymbol{p})\}_{\ell=1}^{L}$ where:

$$f_\ell(\boldsymbol{p}) = \frac{p_\ell + \varepsilon + \max\{0, \min\{z_\ell(\boldsymbol{p}), 1\}\}}{1 + L\varepsilon + \sum_{k=1}^{L} \max\{0, \min\{z_k(\boldsymbol{p}), 1\}\}}$$

- $\sum_{\ell=1}^{L} f_\ell(\boldsymbol{p}) = 1$ and $f_\ell(\boldsymbol{p}) \geq \frac{\varepsilon}{1+2L}\ \forall \ell$
  - $\Rightarrow \boldsymbol{f}(\boldsymbol{p}) \in \Delta_\varepsilon$ for any $\boldsymbol{p} \in \Delta_\varepsilon$. The function maps Δ_ε onto itself.
- Each f_ℓ is also continuous, by the continuity of each z_ℓ and the denominator being bounded away from 0.
- $\boldsymbol{f}(\boldsymbol{p})$ is a continuous function mapping a compact, convex, non-empty set onto itself, so $\exists \boldsymbol{p}^\star$ s.t. $f\left(\boldsymbol{p}^\star\right) = \boldsymbol{p}^\star$.

Letting $\boldsymbol{\varepsilon} \longrightarrow \mathbf{0}$

- Now let $\varepsilon \longrightarrow 0$ and consider the associated sequence of fixed point price vectors $\boldsymbol{p}^n \longrightarrow \boldsymbol{p}$.
- The sequence $\boldsymbol{p}^n \in \mathbb{R}^L$ is bounded because $\boldsymbol{p}^n \in \Delta_\varepsilon \; \forall n$.
- Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence (Bolzano-Weierstrass theorem).
- Call the converged vector $\boldsymbol{p}^\star$.
- Because $\boldsymbol{p}^\star$ is in the simplex, $\boldsymbol{p}^\star \geq \mathbf{0}$ and $\boldsymbol{p} \neq \mathbf{0}$. We need to show that in fact $\boldsymbol{p}^\star \gg \mathbf{0}$.

Proving that $\boldsymbol{p}^\star \gg \mathbf{0}$

Because $f(\boldsymbol{p}^n) = \boldsymbol{p}^n$, every price vector in the sequence satisfies $(\forall \ell)$:

$$p_\ell^n \left[1 + L\varepsilon + \sum_{k=1}^{L} \max\{0, \min\{z_k(\boldsymbol{p}^n), 1\}\} \right] = p_\ell^n + \varepsilon + \max\{0, \min\{z_\ell(\boldsymbol{p}^n), 1\}\}$$

- Suppose $p_k^\star = 0$ for some good k. Then, as $p_k^n \to 0$:

$$\underbrace{p_k^n}_{\to 0} \underbrace{\left[L\varepsilon + \sum_{m=1}^{L} \max\{0, \min\{z_m(\boldsymbol{p}^n), 1\}\} \right]}_{\text{Positive, by property (v) and bounded due to the min}} =$$

$$\underbrace{\varepsilon}_{\to 0} + \underbrace{\max\{0, \min\{z_k(\boldsymbol{p}^n), 1\}\}}_{=1\text{, by property (v)}}$$

- LHS$\to 0$ but RHS$\to 1$. Therefore it must be that $\boldsymbol{p}^\star \gg \mathbf{0}$.

Last Step: Show that $\boldsymbol{f}\left(\boldsymbol{p}^{\star}\right)=\boldsymbol{p}^{\star}$ is an Equilibrium

- We now show that $\boldsymbol{f}\left(\boldsymbol{p}^{\star}\right)=\boldsymbol{p}^{\star}$ is an equilibrium ($\boldsymbol{z}\left(\boldsymbol{p}^{\star}\right)=\mathbf{0}$).
- The fixed point condition implies that (after $\varepsilon \longrightarrow 0$):

$$p_{\ell}^{\star}\left[\sum_{k=1}^{L} \max\left\{0, \min\left\{z_k\left(\boldsymbol{p}^{\star}\right), 1\right\}\right\}\right] = \max\left\{0, \min\left\{z_{\ell}\left(\boldsymbol{p}^{\star}\right), 1\right\}\right\}$$

$$\underbrace{\sum_{\ell=1}^{L} z_{\ell}\left(\boldsymbol{p}^{\star}\right) p_{\ell}^{\star}}_{=0 \text{ by Walras' Law}} \underbrace{\left[\sum_{k=1}^{L} \max\left\{0, \min\left\{z_k\left(\boldsymbol{p}^{\star}\right), 1\right\}\right\}\right]}_{\text{Bounded due to the min}} = \sum_{\ell=1}^{L} z_{\ell}\left(\boldsymbol{p}^{\star}\right) \underbrace{\max\left\{0, \min\left\{z_{\ell}\left(\boldsymbol{p}^{\star}\right), 1\right\}\right\}}_{0 \text{ if } z_{\ell}\left(\boldsymbol{p}^{\star}\right)<0}$$

- The LHS is zero, so the RHS must be zero.
  - Can't have any $z_{\ell}\left(\boldsymbol{p}^{\star}\right)>0$ because RHS must sum to zero and no term on the RHS can be negative, so we must have $\boldsymbol{z}\left(\boldsymbol{p}^{\star}\right) \leq \mathbf{0}$.
  - Can't have any $z_{\ell}\left(\boldsymbol{p}^{\star}\right)<0$ when $\boldsymbol{z}\left(\boldsymbol{p}^{\star}\right) \leq \mathbf{0}$ and $\boldsymbol{p}^{\star} \gg \mathbf{0}$ because of Walras' law: $\sum_{\ell=1}^{L} p_{\ell} z_{\ell}\left(\boldsymbol{p}^{\star}\right)=0$.
  - Therefore the RHS is only zero if $\boldsymbol{z}\left(\boldsymbol{p}^{\star}\right)=\mathbf{0}$.

Uniqueness of Walrasian Equilibria

- Certain conditions on preferences and/or the endowments can guarantee that there will be a unique equilibrium:
  1. Strict convexity and Pareto optimality of the initial endowment.
  2. Aggregate excess demand function has the gross substitute property for all goods.
  3. If $\boldsymbol{Dz}(\boldsymbol{p})$ has full rank and is NSD.
- We will consider each of these cases in turn.
- Assume throughout that each consumer's preferences are continuous, strictly convex and strongly monotone and $\boldsymbol{\omega}_i \gg \mathbf{0}$.

Pareto Optimality of the Initial Endowment

Proposition

In a pure exchange economy, if $\boldsymbol{\omega}_i \gg \mathbf{0}$, $X_i = \mathbb{R}^L_+$, and preferences $\succeq_i$ satisfy continuity, strong monotonicity, and strict convexity for all i, then if $(\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_I)$ is Pareto optimal, then $\boldsymbol{x}_i^\star = \boldsymbol{\omega}_i \; \forall i$ is the unique equilibrium allocation.

- $\boldsymbol{x}_i = \boldsymbol{\omega}_i \; \forall i$ is an equilibrium by the 2nd Welfare Theorem.
- Suppose $\boldsymbol{x}' \neq \boldsymbol{\omega}$ and $\boldsymbol{p}'$ is also an equilibrium.
- Because $\boldsymbol{x}'$ is an equilibrium, $\boldsymbol{x}'_i \succeq_i \boldsymbol{\omega}_i \; \forall i$.
- It also satisfies feasibility: $\sum_{i=1}^I \boldsymbol{x}'_i = \sum_{i=1}^I \boldsymbol{\omega}_i$.
- By strict convexity, $\boldsymbol{x}''_i = \frac{1}{2}\boldsymbol{x}'_i + \frac{1}{2}\boldsymbol{\omega}_i$ satisfies $\boldsymbol{x}''_i \succ_i \boldsymbol{\omega}_i \; \forall i$.
- Moreover, $\boldsymbol{x}''_i$ is feasible because:

$$\sum_{i=1}^I \boldsymbol{x}''_i = \frac{1}{2}\sum_{i=1}^I \boldsymbol{x}'_i + \frac{1}{2}\sum_{i=1}^I \boldsymbol{\omega}_i = \sum_{i=1}^I \boldsymbol{\omega}_i$$

- So $\boldsymbol{x}''$ Pareto dominates $\{\boldsymbol{\omega}_i\}_{i=1}^I$, contradicting that it was Pareto optimal.

The Gross Substitute Property

Definition

The function $\boldsymbol{z}(\cdot)$ has the *gross substitute* (GS) property if whenever $\boldsymbol{p}'$ and $\boldsymbol{p}$ are such that, for some ℓ, $p'_\ell > p_\ell$ and $p'_k = p_k$ for $k \neq \ell$, we have $z_k(\boldsymbol{p}') > z_k(\boldsymbol{p})$ for all $k \neq \ell$.

For small changes, the gross substitute property means:

- $\frac{\partial z_k(\boldsymbol{p})}{\partial p_\ell} > 0$ for all $k \neq \ell$.
- This means $\boldsymbol{Dz}(\boldsymbol{p})$ is positive off the diagonal.
- Because $\boldsymbol{z}(\boldsymbol{p})$ is HD0, $\boldsymbol{Dz}(\boldsymbol{p}) \cdot \boldsymbol{p} = \mathbf{0}$, so the diagonal of $\boldsymbol{Dz}(\boldsymbol{p})$ must be negative.

If every individual satisfies GS, then so does aggregate demand.

Two $L = 2$ Pure Exchange Examples

- Cobb-Douglas utility: $u(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha}$, $\alpha \in (0, 1)$.

$$\boldsymbol{z}(\boldsymbol{p}) = \left(\frac{\alpha (p_1\omega_1 + p_2\omega_2)}{p_1} - \omega_1, \frac{(1-\alpha)(p_1\omega_1 + p_2\omega_2)}{p_2} - \omega_2 \right)$$

$$\boldsymbol{D}\boldsymbol{z}(\boldsymbol{p}) = \begin{pmatrix} -\frac{\alpha p_2 \omega_2}{p_1^2} & \frac{\alpha \omega_2}{p_1} \\ \frac{(1-\alpha)\omega_1}{p_2} & -\frac{(1-\alpha) p_1 \omega_1}{p_2^2} \end{pmatrix}$$

  Positive off the diagonal $\Rightarrow$ Satisfies GS property (if $\omega_\ell > 0 \ \forall \ell$).

- Quasilinear utility: $u(x_1, x_2) = x_1 + 2\sqrt{x_2}$, where we assume $\boldsymbol{p} \cdot \boldsymbol{\omega} > 1/p_2^2$.

$$\boldsymbol{z}(\boldsymbol{p}) = \left(\frac{p_2}{p_1}\omega_2 - \frac{1}{p_1 p_2}, \frac{1}{p_2^2} - \omega_2 \right)$$

  $\frac{\partial z_1(\boldsymbol{p})}{\partial p_2} = \frac{\omega_2}{p_1} + \frac{1}{p_1 p_2^2}$ and $\frac{\partial z_2(\boldsymbol{p})}{\partial p_1} = 0 \Rightarrow$ Violates GS property.

GS Implies Uniqueness in Exchange Economies

Proposition

An aggregate excess demand function $\boldsymbol{z}(\cdot)$ that satisfies the gross substitution property has at most one exchange equilibrium.

- Suppose $\boldsymbol{p}$ and $\boldsymbol{p}'$ were both equilibrium price vectors (and $\boldsymbol{p}'$ was not proportional to $\boldsymbol{p}$.)
- We need to show that $\boldsymbol{z}(\boldsymbol{p}) = \boldsymbol{z}(\boldsymbol{p}') = \boldsymbol{0}$ is not possible.
- Let $m = \max_{\ell} \{p'_\ell / p_\ell\}$ (by strong monotonicity, $\boldsymbol{p} \gg \boldsymbol{0}$).
- For at least one good, $p'_k = mp_k$, and $\boldsymbol{z}(m\boldsymbol{p}) = \boldsymbol{0}$ by HD0.
- Now imagine lowering the price of each good $\ell \neq k$ sequentially from mp_ℓ to p'_ℓ.
  - By GS, the demand for good k will never increase.
  - The demand for good k decreases whenever $p'_\ell \neq mp_\ell$.
  - This happens at least once as $\boldsymbol{p}$ and $\boldsymbol{p}'$ are not proportional.

GS Uniqueness Proof with $L = 2$

- Suppose toward a contradiction that (p_1, p_2) and $\left(p_1', p_2'\right)$ where both equilibria with the vectors not proportional.
- Suppose wlog that $\frac{p_2'}{p_2} > \frac{p_1'}{p_1}$.
- Let $p_2' = mp_2$. From above we know that $p_1' < mp_1$.
- Because $\boldsymbol{z}\,(p_1, p_2)$ is HD0, $\boldsymbol{z}\,(mp_1, mp_2) = \mathbf{0}$.
- When we change prices from (mp_1, mp_2) to $\left(p_1', p_2'\right)$:
  - The price of good 2 doesn't change, but the price of good 1 falls.
  - GS implies that the demand for good 2 *decreases*.
  - But this means that $z_2\left(p_1', p_2'\right) < 0$, contradicting that $\left(p_1', p_2'\right)$ was an equilibrium.

Regular Economies

- Assume the $\boldsymbol{z}(\boldsymbol{p})$ satisfies properties (i)-(v) & is continuously differentiable.
- Normalize $p_L = 1$ and define $\widehat{\boldsymbol{z}}(\boldsymbol{p}) = (z_1(\boldsymbol{p}), \ldots, z_{L-1}(\boldsymbol{p}))$
- With this, $\boldsymbol{p} = (p_1, \ldots, p_{L-1}, 1)$ constitutes a Walrasian equilibrium iff $\widehat{\boldsymbol{z}}(\boldsymbol{p}) = \mathbf{0}$.

Definition

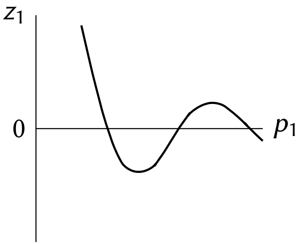
An equilibrium price vector $\boldsymbol{p}$ is *regular* if the $(L-1) \times (L-1)$ matrix of price effects $\boldsymbol{D}\widehat{\boldsymbol{z}}(\boldsymbol{p})$ is nonsingular.

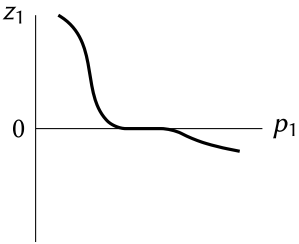
Definition

If every normalized equilibrium price vector is regular, we say that the *economy is regular.*

Regular and Irregular Economies with $L = 2$

If $L = 2$, $\boldsymbol{D}\widehat{\boldsymbol{z}}\,(\boldsymbol{p})$ nonsingular $\Leftrightarrow \frac{\partial z_1(\boldsymbol{p})}{\partial p_1} \neq 0$





- $\frac{\partial z_1(\boldsymbol{p})}{\partial p_1} \neq 0$ at all equilibria
- Each equilibrium is regular
- Economy is regular
- All equilibria are locally isolated
- Finite (odd) number of equilibria

- $\frac{\partial z_1(\boldsymbol{p})}{\partial p_1} = 0$ at all equilibria
- No equilibrium is regular
- Economy is not regular
- No equilibrium is locally isolated
- Infinite number of equilibria

Index Analysis

Definition

Suppose that $\boldsymbol{p} = (p_1, \ldots, p_{L-1}, 1)$ is a regular equilibrium of the economy. Then we denote:

$$\text{index}\left(\boldsymbol{p}\right) = (-1)^{L-1}\,\text{sgn}\left(\left|\boldsymbol{D}\widehat{\boldsymbol{z}}\left(\boldsymbol{p}\right)\right|\right)$$

where $\text{sgn}\left(x\right) = \begin{cases} +1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$

In the left $L = 2$ example, the indices are $+1$, -1, and $+1$

The Index Theorem

For any regular economy, we have:

$$\sum_{\left\{\boldsymbol{p}\in\mathbb{R}^L_+ : \boldsymbol{z}(\boldsymbol{p})=\boldsymbol{0}, p_L=1\right\}} \text{index}\left(\boldsymbol{p}\right) = 1$$

Index Analysis

- For regular economies, the number of equilibria is always odd.
- If $|\boldsymbol{D}\widehat{\boldsymbol{z}}(\boldsymbol{p})| < 0$ at all equilibria, then the equilibrium will be unique.
- The gross substitutes case is a special case of this:
  - $\boldsymbol{Dz}(\boldsymbol{p})$ is NSD whenever $\boldsymbol{z}(\boldsymbol{p}) = \mathbf{0}$ and has rank $L-1$. Therefore the determinant is negative, so its index is $+1$.
- Finally, it can be shown that *almost every* vector of initial endowments $(\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_I) \in \mathbb{R}^{LI}_{++}$, the economy defined by $\{\succeq_i, \boldsymbol{\omega}_i\}_{i=1}^{I}$ is regular.

Gérard Debreu (1921-2004)

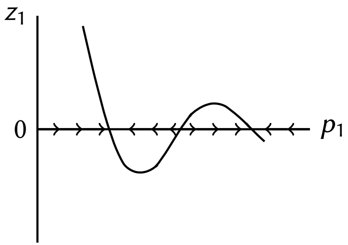
- Born in Calais, France and educated at the École Normale Supérieure Paris Sciences et Lettres.
- Along with his famous work with Kenneth Arrow mentioned above, showed that regular economics have a finite and odd number of price equilibria.
- Won the Nobel Memorial Prize in 1983.

Stability: Price Tâtonnement

- Suppose at $t = 0$, the economy is out of equilibrium: $\boldsymbol{z}(\boldsymbol{p}) \neq \mathbf{0}$.
- Assume prices adjust over time according to:

$$\frac{dp_\ell}{dt} = c_\ell z_\ell(\boldsymbol{p}) \quad \forall \ell$$

  where $c_\ell > 0$ is the speed of adjustment.
- Example with $L = 2$:



Local and System Stability when $L = 2$

- Equilibrium relative prices $\frac{\bar{p}_1}{\bar{p}_2}$ are *locally stable* if, when $\frac{p_1(0)}{p_2(0)}$ is close to it, the dynamic trajectory causes relative prices to converge to $\frac{\bar{p}_1}{\bar{p}_2}$.
- Conversely, equilibrium relative prices $\frac{\bar{p}_1}{\bar{p}_2}$ are *locally totally unstable* if relative prices to diverge from $\frac{\bar{p}_1}{\bar{p}_2}$.
- If the excess demand function is downward-sloping at $\frac{\bar{p}_1}{\bar{p}_2}$ then the equilibrium is locally stable (and locally totally unstable if upward-sloping).
- There is *system stability* if for any initial position $\frac{p_1(0)}{p_2(0)}$, the corresponding trajectory of relative prices $\frac{p_1(t)}{p_2(t)}$ converges to some equilibrium arbitrarily closely as $t \to \infty$.

Normalizing Prices to a Unit Sphere

- Normalize prices such that $\sum_{\ell=1}^{L} p_\ell^2 = 1$
- Assume $c_\ell = c, \forall \ell$.
- As prices adjust, the Euclidian norm of the price vector changes according to:

$$\frac{d}{dt}\left(\sum_{\ell=1}^{L} p_\ell^2\left(t\right)\right) = \sum_{\ell=1}^{L} 2p_\ell\left(t\right)\frac{dp_\ell}{dt} = 2c\sum_{\ell=1}^{L} p_\ell\left(t\right) z_\ell\left(\boldsymbol{p}\right) = 0$$

  where the last equality is from Walras' law.
- Therefore prices are always on the unit sphere as they adjust.

Examples

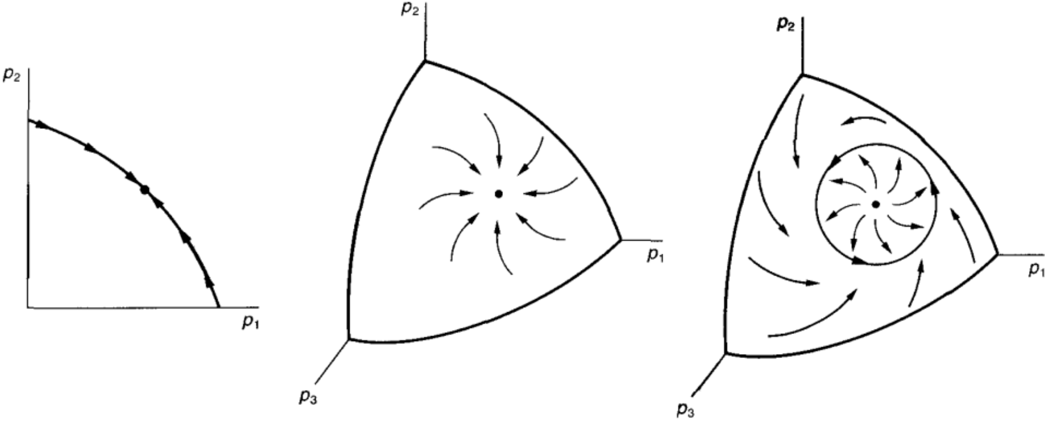


Image Source: Varian, Hal R. (2016) *Microeconomic analysis*

- In the first case, there is a unique stable equilibrium.
- In the second case, there is a unique stable equilibrium.
- In the third case, there is a unique totally unstable equilibrium.

WARP, GS and Globally Stability

- WARP is defined as:

$$\text{If } \boldsymbol{z}(\boldsymbol{p}) \neq \boldsymbol{z}(\boldsymbol{p}') \text{ and } \boldsymbol{p}' \cdot \boldsymbol{z}(\boldsymbol{p}) \leq 0, \text{ then } \boldsymbol{p} \cdot \boldsymbol{z}(\boldsymbol{p}') > 0$$

  So if $\boldsymbol{z}(\boldsymbol{p}) = \boldsymbol{0}$, then $\boldsymbol{p}' \cdot \boldsymbol{z}(\boldsymbol{p}) = 0$, so $\boldsymbol{p} \cdot \boldsymbol{z}(\boldsymbol{p}') > 0$.
- GS $\nRightarrow$ WARP and WARP $\nRightarrow$ GS.
- However, both properties imply the following:

$$\text{If } \boldsymbol{z}(\boldsymbol{p}) = \boldsymbol{0} \text{ and } \boldsymbol{z}(\boldsymbol{p}') \neq \boldsymbol{0}, \text{ then } \boldsymbol{p} \cdot \boldsymbol{z}(\boldsymbol{p}') > 0$$

- GS with $L = 2$, $p_2 = 1$ and $\boldsymbol{z}(\boldsymbol{p}) = \boldsymbol{0}$.
  - GS with $p_1' > p_1$ implies $z_1(\boldsymbol{p}') < z_1(\boldsymbol{p}) = 0$.
  - GS with $p_1' < p_1$ implies $z_1(\boldsymbol{p}') > z_1(\boldsymbol{p}) = 0$.
  - Therefore $(p_1' - p_1)\, z_1(\boldsymbol{p}') < 0$. So:

$$\boldsymbol{p} \cdot \boldsymbol{z}(\boldsymbol{p}') = p_1 z_1(\boldsymbol{p}') + z_2(\boldsymbol{p}') > p_1' z_1(\boldsymbol{p}') + z_2(\boldsymbol{p}') \overset{Walras}{=} 0$$

Global Stability

The following proposition ensures that the GS case we studied in the uniqueness section have a globally stable equilibrium:

Proposition

Suppose that $\boldsymbol{z}\left(\boldsymbol{p}^{\star}\right) = \mathbf{0}$ and $\boldsymbol{p}^{\star} \cdot \boldsymbol{z}\left(\boldsymbol{p}\right) > 0$ for every $\boldsymbol{p}$ not proportional to $\boldsymbol{p}^{\star}$. Then the relative prices of any solution trajectory of the differential equation $\frac{dp_\ell}{dt} = c_\ell z_\ell\left(\boldsymbol{p}\right)$, with $c_\ell > 0 \ \forall \ell$ converge to the relative prices of $\boldsymbol{p}^{\star}$.

Proof

- Construct a Lyapunov function using the Euclidean distance function:

$$V(\boldsymbol{p}) = \sum_{\ell=1}^{L} \frac{1}{c_\ell} \left(p_\ell - p_\ell^\star\right)^2$$

- For $\boldsymbol{p}$ not proportional to $\boldsymbol{p}^\star$:

$$\begin{aligned}\frac{dV(\boldsymbol{p})}{dt} &= 2 \sum_{\ell=1}^{L} \frac{1}{c_\ell} \left(p_\ell(t) - p_\ell^\star\right) \frac{dp_\ell(t)}{dt} \\ &= 2 \sum_{\ell=1}^{L} \frac{1}{c_\ell} \left(p_\ell(t) - p_\ell^\star\right) c_\ell z_\ell(p(t)) = -2\boldsymbol{p}^\star \cdot \boldsymbol{z}(\boldsymbol{p}(t)) < 0\end{aligned}$$

- Because $\boldsymbol{p}^\star$ minimizes $V(\boldsymbol{p})$ and $\frac{dV(\boldsymbol{p}(t))}{dt} < 0 \; \forall \boldsymbol{p} \neq \boldsymbol{p}^\star$, by Lyapunov's Theorem, $\boldsymbol{p}^\star$ is globally stable.